Maths — Assignment - 1

1. $u = [2, 3, -1]$ $\quad v = [4, -2, 5]$

$u + v = [2+4, 3-2, -1+5] = [6, 1, 4]$

$2u - 3v = [4-12, 6+6, -2-15] = [-8, 12, -17]$

2. $a + kb = [7, -6, 9]$, $a = [1, 0, 4]$ $\quad b = [-3, 2, 1]$

$\Rightarrow kb = [7-1, -6-0, 9-4]$

$\Rightarrow kb = [6, -6, 5]$

$\Rightarrow k \times [-3, 2, 1] = [6, -6, 5]$

$\Rightarrow [-3k, 2k, k] = [6, -6, 5]$

$-3k = 6$, $2k = -6$, $k = 5$

$k = -2$, $k = -3$, $k = 5$

$\therefore$ no solution exists.

3. $w = [5, -12, 0]$

euclidian norm $= \sqrt{5^2 + (-12)^2 + (0)^2}$

$= \sqrt{25 + 144}$

$= \sqrt{169}$

$\|w\| = 13$

4. $p = [3, 4, 12]$

$\|p\| = \sqrt{3^2 + 4^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169}$

$|p| = 13$

$\hat{p} = \frac{p}{|p|} = \left[\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right]$

5. $u = [1, 2, 3]$ $\quad v = [4, 5, 6]$

$\|u\| = \sqrt{1^2+2^2+3^2} = \sqrt{1+4+9} = \sqrt{14}$

$\|v\| = \sqrt{4^2+5^2+6^2} = \sqrt{16+25+36} = \sqrt{77}$

$u+v = [5, 7, 9]$

$\|u+v\| = \sqrt{5^2+7^2+9^2} = \sqrt{25+49+81} = \sqrt{155}$
$= 12.45$

$\|u\| + \|v\| = \sqrt{14} + \sqrt{77} = 3.74 + 8.77 = 12.51$

so $\|u\| + \|v\| > \|u+v\|$.

6. $a = [2, 5, -3]$ $\quad b = [1, -4, 7]$

$a \cdot b = 2.1 + 5.(-4) + (-3).7$

$= 2 - 20 - 21$

$\Rightarrow \boxed{a \cdot b = -39}$

7. $u = [1, 2, 3]$ $\quad v = [3, -1, 0]$

$u \cdot v = 1.3 + 2.(-1) + 3.0$
$= 3 - 2$
$= 1$

to be orthogonal $\cos\theta = 0$ i.e. $\theta = 90$.

$u \cdot v \neq 0$

so they are not orthogonal.

8. $b = [3, 4, 5]$ $\quad a = [1, 1, 1]$ $\quad \text{proj}_a b = ?$

$a \cdot b = 3.1 + 4.1 + 5.1$
$= 3+4+5 = 12$

$a \cdot a = 1^2 + 1^2 + 1^2$
$= 3$

so, $\frac{a \cdot b}{a^2} = \frac{12}{3} = 4$

$\text{proj}_a b = \frac{(a \cdot b)}{a^2} \times a = 4 \times [1, 1, 1] = [4, 4, 4]$.

9. $a = [1,1,1]$ $b = [3,4,5]$

scalar $\text{proj}_a b = \frac{a \cdot b}{\|a\|} = \frac{12}{\sqrt{3}} = 4\sqrt{3}$.

10. $x = [2,3,5]$ $y = [1,4,6]$

cosine similarity $= \frac{x \cdot y}{\|x\| \cdot \|y\|}$

$\|x\| = \sqrt{2^2+3^2+5^2} = \sqrt{4+9+25} = \sqrt{38}$

$\|y\| = \sqrt{1^2+4^2+6^2} = \sqrt{1+16+36} = \sqrt{53}$

$x \cdot y = 2.1 + 3.4 + 5.6 = 2+12+30$
$= 44$

cosine similarity $= \frac{44}{\sqrt{38}\sqrt{53}} = \frac{44}{\sqrt{2014}}$.

11. $u = [1,0,2]$ $v = [0,3,1]$

for orthogonal, we need to find cross product

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & k \\ 1 & 0 & 2 \\ 0 & 3 & 1 \end{vmatrix}$$

$= (0-6)\hat{i} - (1-0)\hat{j} + (3-0)\hat{k}$
$= -6\hat{i} - \hat{j} + 3k$

so, $u \times v = [-6,-1,3]$

12. $a = [1,1,1]$ $b = [1,2,3]$ $c = [2,3,5]$

$a \cdot b = 1.1 + 1.2 + 1.3 = 1+2+3 = 6$
$b \cdot c = 1.2 + 2.3 + 3.5 = 2+6+15 = 23$
$a \cdot c = 1.2 + 1.3 + 1.5 = 2+3+5 = 10$

as all of them are $\neq 0$

so they are not pairwise orthogonal.

13. $S = \{[1,0,0], [0,1,0], [0,0,1]\}$

$a = [1,0,0]$ $\quad b = [0,1,0]$ $\quad c = [0,0,1]$

$a \cdot b = 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 = 0$

$b \cdot c = 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 = 0$

$a \cdot c = 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 1 = 0$

so they are orthogonal

$\|a\| = \sqrt{1^2 + 0^2 + 0^2} = \sqrt{1} = 1$

$\|b\| = \sqrt{0^2 + 1^2 + 0^2} = \sqrt{1} = 1$

$\|c\| = \sqrt{0^2 + 0^2 + 1^2} = \sqrt{1} = 1$

so they are normal

orthogonal + normal $\longrightarrow$ orthonormal.

14. $v_1 = [3,4,0]$ $\quad \|v_1\| = \sqrt{3^2 + 4^2 + 0^2} = 5$

$v_2 = [-4,3,0]$ $\quad \|v_2\| = \sqrt{(-4)^2 + 3^2 + 0^2} = 5$

$v_3 = [0,0,1]$ $\quad \|v_3\| = \sqrt{0^2 + 0^2 + 1^2} = 1$

normal vectors are—

$v_1 = [3/5, 4/5, 0]$ $\quad v_2 = [-4/5, 3/5, 0]$ $\quad v_3 = [0,0,1]$

$v_1 \cdot v_2 = \frac{3}{5} \cdot \frac{-4}{5} + \frac{4}{5} \cdot \frac{3}{5} + 0 \cdot 1 = 0$

$v_2 \cdot v_3 = -\frac{4}{5} \cdot 0 + \frac{3}{5} \cdot 0 + 0 \cdot 1 = 0$

$v_1 \cdot v_3 = \frac{3}{5} \cdot 0 + \frac{4}{5} \cdot 0 + 0 \cdot 1 = 0$

so they are now orthonormal.

15. $v = [-2, 5, -3, 1]$

- L-1 norm

$$\|v\|_1 = |(-2)| + |5| + |(-3)| + |1|$$
$$= 11$$

- L-2 norm

$$\|v\|_2 = \sqrt{(-2)^2 + (5)^2 + (-3)^2 + (1)^2}$$
$$= \sqrt{4 + 25 + 9 + 1}$$
$$= \sqrt{39}.$$

16. $v_1 = [1, 0, 1]$ $\quad v_2 = [0, 1, 1]$

vector $= [1, 2, 3]$

for any $c_1 v_1 + c_2 v_2 =$ vector $[1, 2, 3]$

so, $c_1 = 1 \quad c_2 = 2$

$1[1, 0, 1] + 2[0, 1, 1]$

$= [1, 0, 1] + [0, 2, 2]$

$= [1, 2, 3] \longrightarrow$ vector

so, yes it is the span of v_1 & v_2.

17. $c_1[1, 1, 1] + c_2[2, 3, 4] = [5, 7, 9]$

$\Rightarrow [c_1 + 2c_2, c_1 + 3c_2, c_1 + 4c_2] = [5, 7, 9]$

$c_1 + 2c_2 = 5$ — (1)
$c_1 + 3c_2 = 7$ — (2)
$c_1 + 4c_2 = 9$ — (3)

using (1) & (2):

$$\begin{array}{r} c_1 + 2c_2 = 5 \\ - \; c_1 + 3c_2 = 7 \\ \hline -c_2 = -2 \end{array}$$

$c_2 = 2$

$c_1 + 2.2 = 5$

$c_1 = 1$

verifying —

$1.[1, 1, 1] + 2.[2, 3, 4]$

$= [1+4, 1+6, 1+8]$

$= [5, 7, 9]$ so, $1.[1, 1, 1] + 2.[2, 3, 4] = [5, 7, 9]$.

18. $c_1[1,0,1] + c_2[0,1,1] = [1,1,2]$

for $c_1 = 1$ $c_2 = 1$

$1.[1,0,1] + 1.[0,1,1]$

$\Rightarrow [1+0, 0+1, 1+1]$

$\Rightarrow [1,1,2]$

so it is linearly dependant.

19. $A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 0 & 9 \\ 4 & 7 & 11 \end{bmatrix}$

$$\text{det. } A = \begin{vmatrix} 3 & 1 & 2 \\ 0 & 0 & 9 \\ 4 & 7 & 11 \end{vmatrix}$$

$= 3(0.11 - 9.7) + 1(0.11 - 9.4) + 2(0.7 - 0.4)$

$= 3(-63) + 1(-36) + 2(0)$

$= -189 - 36$

$= -225 \neq 0$

so it is linearly independant.

20. Matrix is non-singular

so, rank is 3.

21. $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 6 \\ 1 & 0 & 1 \end{bmatrix}$

$$\det . A = \begin{vmatrix} 2 & 1 & 3 \\ 4 & 2 & 6 \\ 1 & 0 & 1 \end{vmatrix}$$

$$= 2 . (2.1 - 0) - 1 (4.1 - 6.1) + 3 (4.0 - 1.2)$$

$$= 2 \cdot 2 + 1 . 2 + 3 (-2)$$

$$= 4 + 2 - 6$$

$$= 0$$

so it is a dependant matrix.

also $[4, 2, 6]$ is a multiple of $[2, 1, 3]$

so independant vectors $= \{[2, 1, 3], [1, 0, 1]\}$

22. $v_1 = [1, 2]$ $\quad v_2 = [3, 4]$ $\quad v_3 = [5, 6]$

$$c_1 v_1 + c_2 v_2 = v_3$$

$$\Rightarrow c_1 [1, 2] + c_2 [3, 4] = [5, 6]$$

for $c_1 = 1$ $\quad c_2 = 2$

$$1 . [1, 2] + 2 [3, 4] = [5, 6]$$

$$[6 - 1, 8 - 2] = [5, 6]$$

$$[5, 6] = [5, 6]$$

so they are linearly dependant.

23. $v_1 = [1,1,0]$ $v_2 = [2,2,0]$

$v_3 = [0,0,1]$ $v_4 = [3,3,1]$

v_2 is a multiple of v_1

v_4 is $3v_1 + v_3$.

so let whole is a linearly dependent.

independent = $\{[1,1,0], [0,0,1]\}$

24. $v_1 = [1,2,3]$ $v_2 = [4,5,6]$ $v_3 = [7,8,9]$

$c_1 v_1 + c_2 v_2 = v_3$

$c_1[1,2,3] + c_2[4,5,6] = [7,8,9]$

for $c_1 = 1$ $c_2 = 2$

$1[1,2,3] + 2[4,5,6] = [7,8,9]$

$\Rightarrow [8-1, 10-2, 12-3] = [7,8,9]$

$\Rightarrow [7,8,9] = [7,8,9]$

it is dependent.

so, independent = $\{[1,2,3], [4,5,6]\}$.